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Cutting tool

Abstract

A cutting tool is rotatable about a central axis and has a front end in an axial direction along the central axis. The cutting tool includes a body, a blade, a first attachment screw, an adjustment piece, and a second attachment screw. The body has an outer peripheral surface. A pocket is formed in the outer peripheral surface at an end portion on the front end side. The blade is attached to the body by screwing the first attachment screw into the body in the pocket. The blade has a blade side surface facing inward in a radial direction that is orthogonal to the axial direction and that passes through the central axis. An inner wall surface of the pocket has a contact surface facing the blade side surface with a space being interposed between the blade side surface and the contact surface in the radial direction.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) The present application is based on PCT filing PCT/JP2022/002921, filed Jan. 26, 2022, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a cutting tool.

BACKGROUND ART

(3) Japanese Patent Laying-Open No. 2013-176827 (PTL 1) describes a cutting tool. The cutting tool described in PTL 1 includes a tool main body, a cartridge, a cutting insert, and a vibration adjustment mechanism. The cutting insert is attached to the cartridge. The cartridge is attached to the tool main body. The vibration adjustment mechanism adjusts an amount of protrusion of a cutting edge of the cutting insert in a radial direction.

CITATION LIST

Patent Literature

(4) PTL 1: Japanese Patent Laying-Open No. 2013-176827

SUMMARY OF INVENTION

(5) A cutting tool of the present disclosure is rotatable about a central axis and has a front end in an axial direction along the central axis. The cutting tool includes a body, a blade, a first attachment screw, an adjustment piece, and a second attachment screw. The body has an outer peripheral surface. A pocket is formed in the outer peripheral surface at an end portion on the front end side. The blade is attached to the body by screwing the first attachment screw into the body in the pocket. The blade has a blade side surface facing inward in a radial direction that is orthogonal to the axial direction and that passes through the central axis. An inner wall surface of the pocket has a contact surface facing the blade side surface with a space being interposed between the blade side surface and the contact surface in the radial direction. The space between the blade side surface and the contact surface is smaller in a direction away from the front end in the axial direction. The adjustment piece is disposed between the blade side surface and the contact surface. The second attachment screw is screwed into the body and is advanced or retracted along the axial direction so as to move the adjustment piece along the axial direction.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of a cutting tool **100**.

- (2) FIG. 2 is a front view of cutting tool **100**.
- (3) FIG. 3 is a cross sectional view at III-III in FIG. 2.
- (4) FIG. 4 is a perspective view of a blade **20**.
- (5) FIG. 5 is a front view of an adjustment piece **40**.
- (6) FIG. 6A is a cross sectional view at VIA-VIA in FIG. 5.
- (7) FIG. 6B is a cross sectional view at VIB-VIB in FIG. 5.
- (8) FIG. 6C is a first perspective view of adjustment piece **40**.
- (9) FIG. 6D is a second perspective view of adjustment piece **40**.
- (10) FIG. 7 is a partial plan view of a cutting tool **200**.
- (11) FIG. 8 is a first partial side view of cutting tool **200**.
- (12) FIG. 9 is a second partial side view of cutting tool **200**.
- (13) FIG. 10A is a schematic diagram illustrating a method of measuring an angle variation amount **X1**.
- (14) FIG. 10B is a schematic diagram illustrating a method of measuring an angle variation amount **X2**.
- (15) FIG. 11A is a schematic diagram illustrating a method of measuring a position variation amount **Y1** before and after fully fastening a first attachment screw **30**.
- (16) FIG. 11B is a schematic diagram illustrating a method of measuring a position variation amount **Y2** before and after fully fastening an attachment screw **73**.

DETAILED DESCRIPTION

Problem to be Solved by the Present Disclosure

(17) In the cutting tool described in PTL 1, there is room for improvement in terms of precision in position adjustment of the cutting edge. The present disclosure has been made in view of such a problem of the conventional art. More specifically, the present disclosure provides a cutting tool to improve precision in position adjustment of a cutting edge.

Advantageous Effect of the Present Disclosure

(18) According to the cutting tool of the present disclosure, the precision in the position adjustment of the cutting edge can be improved.

SUMMARY OF EMBODIMENTS

(19) First, embodiments of the present disclosure are listed and described.

(20) (1) A cutting tool according to an embodiment is rotatable about a central axis and has a front end in an axial direction along the central axis. The cutting tool includes a body, a blade, a first attachment screw, an adjustment piece, and a second attachment screw. The body has an outer peripheral surface. A pocket is formed in the outer peripheral surface at an end portion on the front end side. The blade is attached to the body by screwing a first attachment screw into the body in the pocket. The blade has a blade side surface facing inward in a radial direction that is orthogonal to the axial direction and that passes through the central axis. An inner wall surface of the pocket has a contact surface facing the blade side surface with a space being interposed between the blade side surface and the contact surface in the radial direction. The space between the blade side surface and the contact surface is smaller in a direction away from the front end in the axial direction. The adjustment piece is disposed between the blade side surface and the contact surface. The second attachment screw is screwed into the body and is advanced or retracted along the axial direction so as to move the adjustment piece along the axial direction. According to the cutting tool of (1), precision in position adjustment of the cutting edge can be improved.

(21) (2) In the cutting tool of (1), the adjustment piece may have a first piece side surface that is in contact with the blade side surface, and a second piece side surface that is opposite to the first piece side surface in the radial direction and that is in contact with the contact surface. The second piece side surface may be parallel to the contact surface. According to the cutting tool of (2), the precision in position adjustment of the cutting edge can be further improved.

(22) (3) In the cutting tool of (1) or (2), the second attachment screw may have a head portion and a

stem portion connected to the head portion and screwed into the body. The head portion may be in contact with the adjustment piece such that a distance in the radial direction between a center of the stem portion in the radial direction and a center of the adjustment piece in the radial direction is 1.5 mm or less. According to the cutting tool of (3), axial force of the second attachment screw can be facilitated to be transmitted to the adjustment piece, thereby stabilizing movement of the adjustment piece.

(23) (4) In the cutting tool of (1), the adjustment piece may have a first piece side surface that is in contact with the blade side surface, and a second piece side surface that is opposite to the first piece side surface in the radial direction and that is in contact with the contact surface. The second piece side surface may be parallel to the contact surface. The second attachment screw may have a head portion and a stem portion connected to the head portion and screwed into the body. The head portion may be in contact with the adjustment piece such that a distance in the radial direction between a center of the stem portion in the radial direction and a center of the adjustment piece in the radial direction is 1.5 mm or less.

Details of Embodiments

(24) Next, details of the embodiments of the present disclosure will be described with reference to figures. In the below-described figures, the same or corresponding portions are denoted by the same reference characters, and the same explanation will not be described repeatedly. A cutting tool according to an embodiment will be referred to as “cutting tool **100**”.

(25) (Configuration of Cutting Tool **100**)

(26) A configuration of cutting tool **100** will be described below.

(27) FIG. **1** is a perspective view of cutting tool **100**. FIG. **2** is a front view of cutting tool **100**. FIG. **3** is a cross sectional view at III-III in FIG. **2**. FIG. **4** is a perspective view of a blade **20**. FIG. **5** is a front view of an adjustment piece **40**. FIG. **6A** is a cross sectional view at VIA-VIA in FIG. **5**. FIG. **6B** is a cross sectional view at VIB-VIB in FIG. **5**. FIG. **6C** is a first perspective view of adjustment piece **40**. FIG. **6D** is a second perspective view of adjustment piece **40**. FIG. **6D** is a perspective view of adjustment piece **40** when viewed in a direction different from that in FIG. **6C**. As shown in FIGS. **1** to **6D**, cutting tool **100** is a boring quill. However, cutting tool **100** is not limited thereto.

(28) The central axis of cutting tool **100** is defined as a central axis A. A direction along central axis A is defined as an axial direction. A direction orthogonal to the axial direction and passing through central axis A is defined as a radial direction. When viewed along the axial direction, a direction along a circumference centered on central axis A is defined as a peripheral direction.

(29) Cutting tool **100** has a front end **100a** and a base end **100b**. Front end **100a** and base end **100b** are respective ends of cutting tool **100** in the axial direction. Base end **100b** is an end opposite to front end **100a**. Cutting tool **100** is attached to a main shaft of a machine tool on the base end **100b** side. The machine tool rotates cutting tool **100** about central axis A. Cutting tool **100** has a body **10**, blade **20**, a first attachment screw **30**, adjustment piece **40**, a second attachment screw **50**, and an adjustment screw **60**.

(30) <Body **10**>

(31) Body **10** is composed of, for example, steel. Body **10** has an outer peripheral surface **10a**. A plurality of pockets **10b** are formed in outer peripheral surface **10a** at an end portion of body **10** on the front end **100a** side. The plurality of pockets **10b** are formed, for example, at equal intervals in the peripheral direction. Each of pockets **10b** has a seating surface **10ba** and a contact surface **10bb**. Seating surface **10ba** is a portion of an inner wall surface of pocket **10b** in contact with a bottom surface **22b** described later. Contact surface **10bb** is a portion of pocket **10b** facing a below-described first blade side surface **22c** with a space being interposed therebetween in the radial direction in a state in which blade **20** is attached to body **10**. Although not shown, a screw hole is formed in seating surface **10ba**.

(32) <Blade **20**>

(33) Blade **20** has a cutting edge **21**. When cutting edge **21** is brought into contact with a workpiece with cutting tool **100** being rotated about central axis A, a cutting process on the workpiece is performed.

(34) Cutting edge **21** has a first cutting edge **21a**, a second cutting edge **21b**, and a corner cutting edge **21c**. First cutting edge **21a** and second cutting edge **21b** are connected to each other by corner cutting edge **21c**. First cutting edge **21a** extends along the axial direction when attached to body **10**. Second cutting edge **21b** extends along the radial direction when attached to body **10**. Blade **20** has a base body **22** and a cutting edge tip **23**. It should be noted that blade **20** may be constituted only of base body **22**.

(35) Base body **22** is composed of, for example, a cemented carbide. Base body **22** has an upper surface **22a**, a bottom surface **22b**, a first blade side surface **22c**, a second blade side surface **22d**, a third blade side surface **22e**, and a fourth blade side surface **22f**. Upper surface **22a** and bottom surface **22b** are end surfaces in the thickness direction of base body **22**. Bottom surface **22b** is a surface opposite to upper surface **22a**.

(36) Each of first blade side surface **22c**, second blade side surface **22d**, third blade side surface **22e**, and fourth blade side surface **22f** is contiguous to upper surface **22a** at the upper end, and is contiguous to bottom surface **22b** at the lower end. First blade side surface **22c** faces inward in the radial direction when attached to body **10**. Second blade side surface **22d** is a surface opposite to first blade side surface **22c**. Third blade side surface **22e** faces the front end **100a** side when attached to body **10**. Fourth blade side surface **22f** is a surface opposite to third blade side surface **22e**.

(37) Upper surface **22a** has an attachment surface **22aa**. Attachment surface **22aa** is located at an end portion of upper surface **22a** on the third blade side surface **22e** side. A distance between attachment surface **22aa** and bottom surface **22b** is smaller than a distance between a portion of upper surface **22a** other than attachment surface **22aa** and bottom surface **22b**. As a result, a step is formed between attachment surface **22aa** and the portion of upper surface **22a** other than attachment surface **22aa**.

(38) A through hole **22g** is formed in base body **22**. Through hole **22g** extends through base body **22** along the thickness direction. From another viewpoint, it can be said that through hole **22g** is opened at upper surface **22a** and bottom surface **22b**. As described above, first blade side surface **22c** faces contact surface **10bb** with a space being interposed therebetween in the radial direction. The space between first blade side surface **22c** and contact surface **10bb** is smaller in a direction away from front end **100a**.

(39) Cutting edge tip **23** is composed of, for example, a sintered material of cubic boron nitride grains. Cutting edge tip **23** may be composed of a sintered material of diamond grains. However, the constituent material of cutting edge tip **23** is not limited thereto. Cutting edge tip **23** is attached to base body **22**. More specifically, cutting edge tip **23** is attached to attachment surface **22aa** by brazing, for example. Cutting edge **21** is formed in cutting edge tip **23**. It should be noted that when blade **20** does not have cutting edge tip **23**, cutting edge **21** is formed in base body **22**.

(40) <First Attachment Screw **30**>

(41) First attachment screw **30** is used to attach blade **20** to body **10**. More specifically, blade **20** is attached to body **10** in pocket **10b** by inserting first attachment screw **30** into through hole **22g** and screwing first attachment screw **30** into a screw hole formed in seating surface **10ba** with bottom surface **22b** being in contact with seating surface **10ba**.

(42) <Adjustment Piece **40**>

(43) Adjustment piece **40** is used to adjust the position of blade **20** (cutting edge **21**) in the radial direction. Adjustment piece **40** is disposed between first blade side surface **22c** and contact surface **10bb**, and is in contact with first blade side surface **22c** and contact surface **10bb**.

(44) Adjustment piece **40** has a first piece side surface **40a** and a second piece side surface **40b**. First piece side surface **40a** and second piece side surface **40b** are end surfaces of adjustment piece

40 in the radial direction. First piece side surface **40a** is in contact with first blade side surface **22c**. Second piece side surface **40b** is a surface opposite to first piece side surface **40a** and is in contact with contact surface **10bb**. Second piece side surface **40b** is preferably parallel to contact surface **10bb**. As described above, since the space between first blade side surface **22c** and contact surface **10bb** is smaller in the direction away from front end **100a**, when adjustment piece **40** is moved away from front end **100a** along the axial direction, blade **20** is moved outward in the radial direction, with the result that the position of first cutting edge **21a** in the radial direction is moved outward.

(45) Adjustment piece **40** has a third piece side surface **40c** and a fourth piece side surface **40d**. Third piece side surface **40c** and fourth piece side surface **40d** are end surfaces of adjustment piece **40** in the axial direction. Third piece side surface **40c** faces the front end **100a** side. Fourth piece side surface **40d** is a surface opposite to third piece side surface **40c**. Adjustment piece **40** has a first piece side wall **41** and a second piece side wall **42**. First piece side wall **41** and second piece side wall **42** are side walls of adjustment piece **40** that respectively form third piece side surface **40c** and fourth piece side surface **40d**. First piece side wall **41** and second piece side wall **42** face each other with a space being interposed therebetween in the axial direction.

(46) A first through hole **41a** is formed in first piece side wall **41**. First through hole **41a** extends through first piece side wall **41** along the thickness direction (axial direction). One end of first through hole **41a** is opened in third piece side surface **40c**, and the other end of first through hole **41a** is opened in the inside of adjustment piece **40**. A second through hole **42a** is formed in second piece side wall **42**. Second through hole **42a** extends through second piece side wall **42** along the thickness direction (axial direction). One end of second through hole **42a** is opened in fourth piece side surface **40d**, and the other end of second through hole **42a** is opened in the inside of adjustment piece **40**.

(47) The center of adjustment piece **40** in the radial direction is defined as a first position **P1**. First position **P1** is a midpoint between the end of first piece side surface **40a** on the third piece side surface **40c** side and the end of second piece side surface **40b** on the third piece side surface **40c** side. The center of a stem portion **52** (described later) in the radial direction is defined as a second position **P2**. A distance between first position **P1** and second position **P2** in the radial direction is defined as a distance **DIS**. Second position **P2** may be close to first piece side surface **40a** with respect to first position **P1**, or may be close to second piece side surface **40b** with respect to first position **P1**.

(48) <Second Attachment Screw **50**>

(49) Second attachment screw **50** is used to advance or retract adjustment piece **40** along the axial direction. Second attachment screw **50** has a head portion **51** and stem portion **52**. Head portion **51** is disposed inside adjustment piece **40**. Head portion **51** is preferably in contact with the inner wall surface of second piece side wall **42** such that distance **DIS** is 1.5 mm or less. Distance **DIS** is more preferably 0 mm (first position **P1** coincides with second position **P2**).

(50) Stem portion **52** is connected to head portion **51**. Stem portion **52** extends along the axial direction. Stem portion **52** is inserted into second through hole **42a** and is screwed into a screw hole **10c**. It should be noted that screw hole **10c** is formed in body **10** and extends along the axial direction. By inserting the tool from first through hole **41a** and rotating head portion **51** about the central axis of second attachment screw **50**, second attachment screw **50** is advanced or retracted along the axial direction. Since head portion **51** is in contact with the inner wall surface of second piece side wall **42** as described above, adjustment piece **40** is also advanced or retracted along the axial direction in response to second attachment screw **50** being advanced or retracted along the axial direction.

(51) <Adjustment Screw **60**>

(52) An adjustment screw **60** is used to adjust a position of blade **20** (cutting edge **21**) in the axial direction. Adjustment screw **60** has a head portion **61** and a stem portion **62**. Head portion **61** is in

contact with fourth blade side surface **22f**. Stem portion **62** is connected to head portion **61**. Stem portion **62** extends along the axial direction. Stem portion **62** is screwed into a screw hole **10d**. It should be noted that screw hole **10d** is formed in body **10** and extends along the axial direction. Since head portion **61** is in contact with fourth blade side surface **22f**, the position of blade **20** in the axial direction is adjusted by rotating head portion **61** about the central axis of adjustment screw **60** so as to advance or retract adjustment screw **60** along the axial direction.

(53) (Method of Adjusting Position of Blade **20** in Cutting Tool **100**)

(54) A method of adjusting the position of blade **20** in cutting tool **100** will be described below.

(55) First, when adjusting the position of blade **20** in cutting tool **100**, blade **20** is temporarily fastened to body **10** using first attachment screw **30**. A state in which blade **20** is temporarily fastened to body **10** by first attachment screw **30** is a state in which fastening torque of first attachment screw **30** is about 30% of recommended fastening torque. The recommended fastening torque is determined by the screw size of first attachment screw **30**.

(56) Second, the position of blade **20** in the axial direction is adjusted by rotating adjustment screw **60**. Third, the position of blade **20** in the radial direction is adjusted by rotating second attachment screw **50**. Fourth, blade **20** is fully fastened to body **10** by further rotating first attachment screw **30**. A state in which blade **20** is fully fastened to body **10** by first attachment screw **30** is a state in which fastening torque of first attachment screw **30** is more than or equal to the recommended fastening torque. In this way, in cutting tool **100**, the positions of blade **20** in the radial direction and the axial direction are adjusted.

(57) (Effects of Cutting Tool **100**)

(58) Effects of cutting tool **100** will be described below in comparison with a cutting tool according to a comparative example. The cutting tool according to the comparative example will be referred to as “cutting tool **200**”.

(59) FIG. **7** is a partial plan view of cutting tool **200**. FIG. **8** is a first partial side view of cutting tool **200**. FIG. **9** is a second partial side view of cutting tool **200**. In FIGS. **7** to **9**, body **10** is not illustrated. As shown in FIGS. **7** to **9**, in cutting tool **200**, a cartridge **70**, a radial-direction adjustment screw **80**, and an axial-direction adjustment screw **90** are used instead of blade **20**, first attachment screw **30**, adjustment piece **40**, second attachment screw **50**, and adjustment screw **60**. Cartridge **70** has a cartridge main body **71**, a cutting insert **72**, an attachment screw **73**, and an attachment screw **74**.

(60) Cartridge main body **71** (cartridge **70**) extends along the axial direction. Respective ends of cartridge main body **71** in the axial direction are defined as a first end **71a** and a second end **71b**. First end **71a** is an end of cutting tool **200** on the front end side. Second end **71b** is an end opposite to first end **71a**.

(61) A through hole **71c** and a screw hole **71d** are formed in cartridge main body **71**. Through hole **71c** is located close to second end **71b** with respect to the center of cartridge main body **71** in the axial direction. Screw hole **71d** is located close to first end **71a** with respect to the center of cartridge main body **71** in the axial direction. Cartridge main body **71** is attached to body **10** by inserting attachment screw **73** into through hole **71c** and screwing attachment screw **73** into a screw hole (not shown) formed in body **10**.

(62) A through hole **72a** is formed in cutting insert **72**. Cutting insert **72** is attached to cartridge main body **71** by inserting attachment screw **74** into through hole **72a** and screwing attachment screw **74** into a screw hole (not shown) formed in cartridge main body **71**. Cutting insert **72** is attached to an end portion of cartridge main body **71** on the first end **71a** side.

(63) Cutting insert **72** has a cutting edge **72b**. Cutting edge **72b** has a plurality of straight cutting edges **72ba** and a plurality of corner cutting edges **72bb**. Cartridge **70** is attached to body **10** such that the extending direction of one of the plurality of straight cutting edges **72ba** is along the radial direction in a state in which cartridge main body **71** is attached to body **10**.

(64) Radial-direction adjustment screw **80** is screwed into screw hole **71d**. When radial-direction

adjustment screw **80** is rotated about the central axis, the front end of the stem portion thereof protrudes from cartridge main body **71** and comes into contact with body **10**. Thus, the position of the cutting edge of cutting insert **72** in the radial direction is adjusted. The stem portion of axial-direction adjustment screw **90** is screwed into a screw hole (not shown) formed in an end surface of cartridge main body **71** on the second end **71b** side. The head portion of axial-direction adjustment screw **90** is in contact with body **10**. Therefore, by rotating axial-direction adjustment screw **90** about the central axis, the position of the cutting edge of cutting insert **72** in the axial direction is adjusted.

(65) First, when adjusting the position of cutting insert **72** in cutting tool **200**, cartridge main body **71** is temporarily fastened to body **10** using attachment screw **73**. A state in which cartridge main body **71** is temporarily fastened to body **10** by attachment screw **73** is a state in which fastening torque of attachment screw **73** is about 30% of recommended fastening torque. The recommended fastening torque is determined by the screw size of attachment screw **73**. Second, the position of cutting insert **72** in the axial direction is adjusted by advancing or retracting axial-direction adjustment screw **90** along the axial direction. Third, the position of cutting insert **72** in the radial direction is adjusted by rotating radial-direction adjustment screw **80** to change an amount of protrusion from cartridge main body **71**.

(66) Fourth, cartridge main body **71** is fully fastened to body **10** by further rotating attachment screw **73**. A state in which cartridge main body **71** is fully fastened to body **10** by attachment screw **73** is a state in which fastening torque of attachment screw **73** is more than or equal to the recommended fastening torque. In this way, in cutting tool **200**, the positions of cutting insert **72** (cutting edge **72b**) in the radial direction and the axial direction are adjusted.

(67) In cutting tool **200**, the position of cutting insert **72** in the radial direction is adjusted by protruding the front end of the stem portion of radial-direction adjustment screw **80** with respect to cartridge main body **71** and bringing the front end into contact with body **10**. Further, the position of radial-direction adjustment screw **80** in the axial direction is greatly separated from the position of attachment screw **73** in the axial direction.

(68) Therefore, when the position of cutting insert **72** in the radial direction is adjusted, cartridge main body **71** is deflected with the vicinity of attachment screw **73** serving as a fulcrum. Due to this deflection, the extending direction of cutting edge **72b** is inclined with respect to the radial direction. Deflection of cartridge main body **71** occurs also when attachment screw **73** is fully fastened. Accordingly, even though the position of cutting insert **72** in the radial direction is adjusted, the position of cutting insert **72** in the radial direction is displaced when fully fastening attachment screw **73**. Thus, in cutting tool **200**, precision in position adjustment of cutting insert **72** (cutting edge **72b**) becomes low.

(69) On the other hand, in cutting tool **100**, the position of blade **20** in the radial direction is adjusted by advancing or retracting adjustment piece **40**, which is located between contact surface **10bb** and first blade side surface **22c**, along the axial direction. The position of adjustment piece **40** in the axial direction is close to the position of first attachment screw **30** in the axial direction.

(70) As a result, in cutting tool **100**, blade **20** is less likely to be deflected with first attachment screw **30** serving as a fulcrum, with the result that second cutting edge **21b** is less likely to be inclined with respect to the radial direction when adjusting the position of blade **20** in the radial direction and the position of blade **20** in the radial direction is less likely to be displaced when fully fastening first attachment screw **30**. Thus, according to cutting tool **100**, precision in position adjustment of blade **20** (cutting edge **21**) can be improved.

(71) When contact surface **10bb** and second piece side surface **40b** are parallel to each other, force is likely to be applied from adjustment piece **40** to blade **20** in a direction close to a direction parallel to the radial direction. Therefore, in this case, second cutting edge **21b** is further less likely to be inclined with respect to the radial direction when adjusting the position of blade **20** in the radial direction, and displacement of the position in the radial direction is further less likely to

occur when fully fastening first attachment screw **30**.

(72) When head portion **51** is in contact with adjustment piece **40** such that distance DIS is 1.5 mm or less, axial force of second attachment screw **50** is facilitated to be transmitted to adjustment piece **40**, thereby stabilizing movement of adjustment piece **40**.

(73) (Evaluation on Precision in Position Adjustment of Cutting Edge)

(74) A first test and a second test were performed in order to evaluate precision in position adjustment of cutting edge **21** in cutting tool **100** and precision in position adjustment of cutting edge **72b** in cutting tool **200**.

(75) <First Test>

(76) In the first test, evaluations were made on a variation amount (angle variation amount X1) of an inclination angle of second cutting edge **21b** with respect to the radial direction before and after moving the position of blade **20** outward in the radial direction and a variation amount (angle variation amount X2) of an inclination angle of straight cutting edge **72ba** with respect to the radial direction before and after moving the position of cutting insert **72** outward in the radial direction. Each of the inclination angle of second cutting edge **21b** with respect to the radial direction before moving the position of blade **20** outward in the radial direction and the inclination angle of straight cutting edge **72ba** with respect to the radial direction before moving the position of cutting insert **72** outward in the radial direction was 0°.

(77) FIG. **10A** is a schematic diagram illustrating a method of measuring angle variation amount X1. As shown in FIG. **10A**, in the measurement of angle variation amount X1, first, the position of blade **20** is moved outward by 0.05 mm in the radial direction by using second attachment screw **50**. Second, the shape of cutting edge **21** is measured using venturion **450** (hereinafter, referred to as “cutting edge measurement instrument”) provided by Zoller. On this occasion, a measurement program used in the cutting edge measurement instrument is Lasso function/No. 21, a measurement mode of the cutting edge measurement instrument is measurement NN, and measurement precision of the cutting edge measurement instrument is 0.001 mm. Third, an angle between second cutting edge **21b** and the radial direction is calculated based on the shape of cutting edge **21** obtained as described above. This angle represents angle variation amount X1.

(78) FIG. **10B** is a schematic diagram illustrating a method of measuring angle variation amount X2. As shown in FIG. **10B**, in the measurement of angle variation amount X2, first, the position of cutting insert **72** is moved outward by 0.05 mm in the radial direction by using radial-direction adjustment screw **80**. Second, the shape of cutting edge **72b** is measured using a cutting edge measurement instrument. Measurement conditions on this occasion are the same as those described above. Third, an angle between straight cutting edge **72ba** and the radial direction is calculated based on the shape of cutting edge **72b** obtained as described above. This angle represents angle variation amount X2.

(79) In cutting tool **200**, angle variation amount X2 was 0.19°. On the other hand, in cutting tool **100**, angle variation amount X1 was 0.03°. That is, in cutting tool **100**, the variation amount of the cutting edge angle before and after moving it outward in the radial direction was reduced by 84% as compared with cutting tool **200**.

(80) <Second Test>

(81) In the second test, evaluations were made on a position variation amount (position variation amount Y1) of blade **20** in the radial direction before and after fully fastening first attachment screw **30** and a position variation amount (position variation amount Y2) of cutting insert **72** in the radial direction before and after fully fastening attachment screw **73**.

(82) FIG. **11A** is a schematic diagram illustrating a method of measuring position variation amount Y1 before and after fully fastening first attachment screw **30**. As shown in FIG. **11A**, in the measurement of the position variation of blade **20** in the radial direction before and after fully fastening first attachment screw **30**, first, the shape of cutting edge **21** before fully fastening first attachment screw **30** (i.e., in a state in which first attachment screw **30** is temporarily fastened) is

measured. Second, first attachment screw **30** is fully fastened. Third, the shape of cutting edge **21** after fully fastening first attachment screw **30** is measured.

(83) Fourth, position variation amount **Y1** is obtained by comparing an outermost position on cutting edge **21** in the radial direction before fully fastening first attachment screw **30** with an outermost position on cutting edge **21** in the radial direction after fully fastening first attachment screw **30**. It should be noted that a method of measuring the shape of cutting edge **21** is the same as that in the first test.

(84) FIG. **11B** is a schematic diagram illustrating a method of measuring position variation amount **Y2** before and after fully fastening attachment screw **73**. As shown in FIG. **11B**, in the measurement of the position variation of cutting insert **72** in the radial direction before and after fully fastening attachment screw **73**, first, the shape of cutting edge **72b** before fully fastening attachment screw **73** (i.e., in a state in which attachment screw **73** is temporarily fastened) is measured. Second, attachment screw **73** is fully fastened. Third, the shape of cutting edge **72b** after fully fastening attachment screw **73** is measured.

(85) Fourth, position variation amount **Y2** is obtained by comparing an outermost position on cutting edge **72b** in the radial direction before fully fastening attachment screw **73** with an outermost position on cutting edge **72b** in the radial direction after fully fastening attachment screw **73**. It should be noted that a method of measuring the shape of cutting edge **72b** is the same as that in the first test.

(86) In cutting tool **200**, position variation amount **Y2** was 0.009 mm. On the other hand, in cutting tool **100**, position variation amount **Y1** was 0.001 mm. That is, in cutting tool **100**, the position variation of the cutting edge in the radial direction before and after fully fastening the attachment screw was reduced by 89% as compared with cutting tool **200**.

(87) The embodiments disclosed herein are illustrative and non-restrictive in any respect. The scope of the present invention is defined by the terms of the claims, rather than the embodiments described above, and is intended to include any modifications within the scope and meaning equivalent to the terms of the claims.

REFERENCE SIGNS LIST

(88) **100, 200**: cutting tool; **100a**: front end; **100b**: base end; **A**: central axis; **10**: body; **10a**: outer peripheral surface; **10b**: pocket; **10ba**: seating surface; **10bb**: contact surface; **10c**: screw hole; **10d**: screw hole; **20**: blade; **21**: cutting edge; **21a**: first cutting edge; **21b**: second cutting edge; **21c**: corner cutting edge; **22**: base body; **22a**: upper surface; **22aa**: attachment surface; **22b**: bottom surface; **22c**: first blade side surface; **22d**: second blade side surface; **22e**: third blade side surface; **22f**: fourth blade side surface; **22g**: through hole; **23**: cutting edge tip; **30**: first attachment screw; **40**: adjustment piece; **40a**: first piece side surface; **40b**: second piece side surface; **40c**: third piece side surface; **40d**: fourth piece side surface; **41**: first piece side wall; **41a**: first through hole; **42**: second piece side wall; **42a**: second through hole; **50**: second attachment screw; **51**: head portion; **52**: stem portion; **60**: adjustment screw; **61**: head portion; **62**: stem portion; **70**: cartridge; **71**: cartridge main body; **71a**: first end; **71b**: second end; **71c**: through hole; **71d**: screw hole; **72**: cutting insert; **72a**: through hole; **72b**: cutting edge; **72ba**: straight cutting edge; **72bb**: corner cutting edge; **73**: attachment screw; **74**: attachment screw; **80**: radial-direction adjustment screw; **90**: axial-direction adjustment screw; **DIS**: distance; **P1**: first position; **P2**: second position; **X1**: angle variation amount; **X2**: angle variation amount; **Y1**: position variation amount; **Y2**: position variation amount.

Claims

1. A cutting tool that is rotatable about a central axis and that has a front end in an axial direction along the central axis, the cutting tool comprising: a body; a blade; a first attachment screw; an adjustment piece; an adjustment screw; and a second attachment screw, wherein the body has an

outer peripheral surface, a pocket is formed in the outer peripheral surface at an end portion on a front end side, the blade is attached to the body by screwing the first attachment screw into the body in the pocket, the blade has a blade side surface facing inward in a radial direction that is orthogonal to the axial direction and that passes through the central axis, an inner wall surface of the pocket has a contact surface facing the blade side surface with a space being interposed between the blade side surface and the contact surface in the radial direction, the space between the blade side surface and the contact surface is smaller in a direction away from the front end in the axial direction, the adjustment piece is disposed between the blade side surface and the contact surface, the second attachment screw is screwed into the body and is advanced or retracted along the axial direction so as to move the adjustment piece along the axial direction, the second attachment screw has a head portion and a stem portion connected to the head portion and screwed into the body, the adjustment piece has a third piece side surface that faces outward in the axial direction, a center of the adjustment piece being a midpoint between opposite edges of the third piece side surface in the radial direction, the adjustment screw extends in parallel with the second attachment screw and is configured to adjust a position of the blade in the axial direction, and the head portion is in contact with the adjustment piece such that a distance in the radial direction between a center of the stem portion in the radial direction and the center of the adjustment piece in the radial direction is 1.5 mm or less.

2. The cutting tool according to claim 1, wherein the adjustment piece has a first piece side surface that is in contact with the blade side surface, and a second piece side surface that is opposite to the first piece side surface in the radial direction and that is in contact with the contact surface, and the second piece side surface is parallel to the contact surface.

3. The cutting tool according to claim 1, wherein the first attachment screw extends lengthwise in a direction orthogonal to the radial direction.
